

IoT-Based Residential Door Access Control System

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0.1 Introduction

This work presents the implementation of a low-cost IoT system for remote control of residential door access. The system uses a Raspberry Pi microcontroller and a webcam to capture and analyze the images for visitor presence detection. When a visitor is detected the system sends an email alert to the homeowner. This integrated platform further allows users to control the door access remotely via a network interface.

0.2 System Architecture and Design Principles

The system is designed as a traditional IoT deployment, consisting of three main layers:

- **Sensing Layer:** This layer is responsible for gathering data from the environment. In our case, the primary sensor is a webcam.
- **Network Layer:** This layer facilitates communication between the system components. Specifically, we use the internet for email messaging for both notification and remote command execution.
- **Application Layer:** The application layer is implemented using a Raspberry Pi, which analyzes sensor readings, generates alarms, and manages the door access system represented by the LED. The email system acts as the user interface for remote access control.

0.2.1 Design Principles

- **Low-Cost and Accessibility:** The system uses readily available, low-cost components such as a Raspberry Pi, USB webcam, and LED.
- **Modularity:** The system's functionalities are decomposed into separate modules for ease of implementation and testing.
- **Remote Accessibility:** The system is remotely observable and controllable through a standard Internet protocol via email.
- **Real-time Operation:** The system captures images and communicates alerts almost instantly, providing timely notifications of visitor presence.

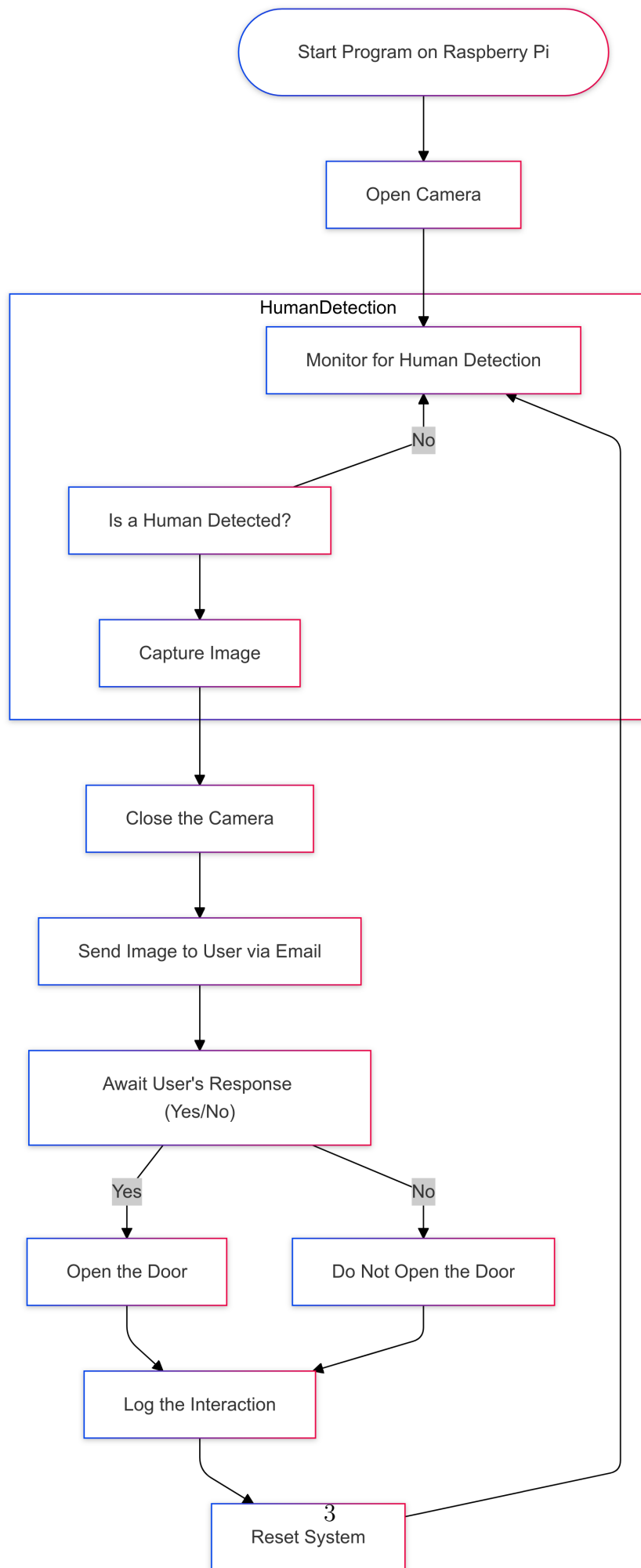


Figure 1 depicts the overall data flow within the system, from sensor input through processing and control.

0.3 Node Types

The system consists of the following types of nodes:

- **Sensor Node:** The webcam acts as a sensor node, processing visual information from the environment. It functions as a continuous sensor.
- **Actuator Node:** The LED acts as an actuator node, controlled by the Raspberry Pi to simulate the door lock status (open/closed).
- **Micro-controller Node:** The Raspberry Pi serves as the microcontroller node, hosting the application code and controlling the sensor and actuator nodes.
- **User Interface:** The email client provides the user interface to control the system remotely.

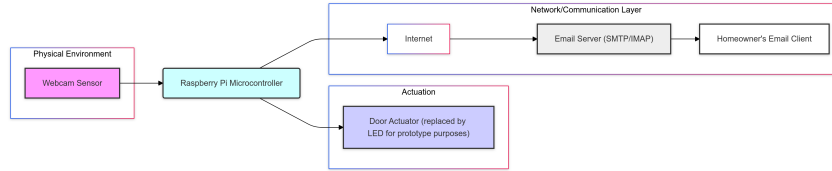


Figure 2: Overall System Structure, Illustrating Node Types

Figure 2 illustrates the overall structure of the system, demonstrating all types of nodes.

0.4 Protocols

The following protocols are used in this system:

- **SMTP (Simple Mail Transfer Protocol):** Used to send email notifications with images to the homeowner.
- **IMAP (Internet Message Access Protocol):** Used to read email responses from the homeowner.
- **GPIO (General Purpose Input/Output):** A hardware-level protocol used to control the LED.
- **COCO dataset:** It is used as a classes reference for the human detection module.

0.5 Modules Description

The developed system is constructed using several software modules, each addressing a specific functionality. The integrated code consists of the following modules:

0.5.1 Email Module

This module handles email communication, including sending alerts and receiving user commands. It is responsible for:

- **Sending Emails:** Using the `smtplib` library, this module sends an email to the user, attaching an image of the detected human and asking for door control permission.
- **Receiving and Parsing Emails:** This module uses the `imaplib` library to constantly check the user's mailbox for answers. It parses the email body to identify whether the user has replied with "yes" or "no". The `email` library is employed to handle email messages, and `pandas` is used for keyword searching.

0.5.2 Human Detection Module

This module focuses on processing visual input to identify human presence using a pre-trained model.

- **Image Capture:** Based on the `cv2` library for obtaining video frames from a connected webcam.
- **Human Detection:** Uses a pre-trained MobileNet model loaded to identify human bounding box in the images captured from the camera.
- **Bounding Box:** Draws a rectangle around the detected human.
- **Image Saving:** Saves the detected human image with a timestamp.
- **Program termination:** The code exits after capturing and saving the image.

0.5.3 LED Control Module

This module is in charge of LED control, which represents the door's lock mechanism.

- **GPIO Control:** Implements an interface with the LED attached to the Raspberry Pi using the `RPi.GPIO` library.
- **LED On/Off Control:** Implements functions to turn the LED on or off, simulating the locking and unlocking of a door.

0.5.4 Integrated System Logic

The overall system operates as follows:

- **Module Integration:** Integrates human detection, email sending/checking, and LED control to achieve the project's functionality.
- **Event Flow Control:** Controls the order of operations based on events, such as detecting human presence, sending email notifications, waiting for a user response, and controlling the LED.

0.6 Application

The IoT-based system operates as follows:

1. **Visitor Detection:** The system continuously monitors the scene using the attached webcam. If the human detection module detects a person, the system proceeds to the next phase.
2. **Image Capture:** Upon detecting a human, the system captures an image of the visitor.
3. **Email Notification:** The captured image is sent as an attachment in an email to the homeowner, prompting them to reply with "yes" or "no" to decide whether to open the door.
4. **User Response:** The system continuously checks the homeowner's email inbox for a response.
5. **Door Control:** Based on the received response:
 - If the homeowner replies with "yes", the LED (simulating the door opening) is turned on.
 - If the homeowner replies with "no", the LED (simulating the door closing) is switched off.

0.7 Conclusion

The resulting system provides a practical, affordable solution for remote residential door control. It combines several IoT peripherals, showcasing basic WSN and IoT concepts. Future enhancements may include:

- **Advanced Object Detection:** Utilizing more advanced models to filter out false positives.
- **Real-time Feedback:** Implementing real-time feedback of the door state via a web browser interface.

This project has demonstrated the potential of low-cost IoT devices in creating practical smart home applications.